OpenRank Visualization Platform: System Architecture

LLM Open-Source Ecosystem Health Analytics Based on OpenDigger Metrics

1 User & Client

- Web Browser
- Student / Analyst / Researcher
- Interactive Dashboard Access

HTTPS Request

Rendered Views

2 Vue 3 Frontend

- Vite Build System
- Vue Router
- Pinia State Management
- API Service Layer
- ECharts Visualization Engine

Main Views

- Home Dashboard
- LLM Ecosystem Summary
- Visual Lab
- Favorites
- Login / Register

Reusable Components

- Metric Line Chart
- Radar Compare
- Contributor Bar Chart

Receives JSON Data from Backend → Renders Charts Dashboard Views

REST API Requests (JSON)

JSON Responses Data Reports AI Insights Favorites

3 Flask Backend Service

Application Entry: main.py

REST APIs

OpenDigger API Module	/api/llm/summary
LLM Summary / Report API	/api/llm/rank/metrics
Contributor Risk API	/api/llm/report
Auth API	/api/auth/login
Auth API (Register)	/api/auth/register
Favorites API	/api/favorites
Contributor Risk API	/api/health/contributors/risk

Middleware / Services

- JWT Authentication
- API Rate Limiting
- CORS
- Request Cache

Process Requests

Cached Data DB Results

Query / Read (Results)

4 Data Processing & Storage

Data Processing Pipeline

- Data Fetcher
- Metric Utilities
- Config Loader
- Health Score Calculation
- Time-Series Aggregation
- Data Cache

Health Score =
0.5 x OpenRank + 0.3 x Activity + 0.2 x Stability

Processed Metrics (Parsed)

SQLite Database

- User
- Favorite
- MetricSeries

Fetch Data (API/Scrape)

5 External Data Sources

- OpenDigger
- OpenRank Metrics
- GitHub Repository Metadata
- Project Activity Metrics
- Developer / Contributor Metrics

Metric Ingestion

- OpenRank
- Activity
- Stars
- Forks
- Issues
- Pull Requests
- Contributors

Platform Capabilities

- Interactive Visualization
- AI Interpretation
- PNG Export
- Report Export
- Favorites Management
- Risk Analysis
- Health Ranking
- Multi-chart Visual Analytics

Legend: Infrastructure / Backend · Data Processing · Analytics / AI Interpretation · Visualization / Capabilities

Primary Data Flow · Response / Return Flow · Bidirectional / Data Access